Return periods and exceedance probabilities

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Summary

Treating a hazard as a yearly coin flip, and what follows from that.

- A Bernoulli process needs one flip per year, independence, and the same probability every year.
- The wait between events is geometric, and the return period is its mean, $T = 1/p$.
- The wait is memoryless, so a basin is no safer the year after a flood.
- The spread of the wait is about as large as the mean wait itself.
- Risk over a design life is one minus the probability of no events, which the binomial gives.
- A return level inverts the distribution: x_T is the level exceeded with probability $1/T$ in a year.
- With a record instead of a distribution, rank the values and use $\hat{p}_i = i/(n+1)$.

Key results.

$$\tau \sim \text{Geometric}(p), \qquad T = E[\tau] = \frac{1}{p}$$

$$R = 1 - (1-p)^N$$

$$x_T = F_X^{-1}\!\left(1 - \frac{1}{T}\right)$$

The two worked cases were a house in the 100-year floodplain over a 30-year mortgage, and a cofferdam sized to a stated 10% risk over 5 years. The longest return period an n-year record can estimate is $n + 1$ years.

Bibliography